

PRATIKSHA SHRESTHA

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SUMMARY

Software developer with 6+ years of experience in full stack development, building scalable web applications and RESTful APIs using C#, ASP.NET Core, Node.js, AWS, and modern JavaScript frameworks. Expertise in microservices and serverless architecture, SQL query optimization, and performance tuning to deliver high-performing applications.

TECHNICAL SKILLS

Languages and Runtime Env: C#, Node.js, Python, JavaScript, TypeScript, PHP, Solidity, Shell Script
Frameworks: React.js, Blazor, ASP.Net Core, Entity Framework, Express.js, Laravel, jQuery, Bootstrap
Databases: MySQL, PostgreSQL, MongoDB, DynamoDB, Oracle Autonomous Database, Redis
Cloud and DevOps: Docker, Linux, AWS, Azure, Azure Pipelines, Bitbucket Pipelines, AWS CodeDeploy
Testing Tools: Selenium, xUnit, Jest, Chai, Hardhat, Foundry
Software/Technology: Microservices, Git, CI/CD, TDD, DRY, MVC, REST API, SOAP, WCF, DApp, Ajax, Nginx, Agile, Jira

EXPERIENCE

Software Engineer | *Communications Test Design Inc., PA* Sep 2024 – Present

- Developed and enhanced warehouse management applications to optimize inventory processes, including stock distribution, returns, and repair workflows - resulting in improved operational throughput.
- Engineered asynchronous **background file upload service** using **.NET hosted services and channels**, significantly reducing processing time and increasing system reliability for large file handling.
- Created and integrated dynamic, carrier-compliant **shipping label generators**, ensuring compliance across carriers.
- Automated logistics workflows by integrating third-party **REST APIs** and consuming **WCF services**, enabling real-time label creation, electronic manifest submission, and SPS-compliant data exchanges.

Tech Stack: C#, ASP.Net Core, Blazor, Entity Framework, PostgreSQL, AWS, Docker, xUnit, TeamCity, Zpl, WCF

Full-Stack Developer | *Myaamia Center, Miami University, OH* Aug 2022 – Aug 2024

- Designed and built intuitive web admin panel for language revitalization, enhancing engagement and accessibility.
- Led **database architecture design** and development for a robust survey management and analysis platform, supporting 10+ survey types with **advanced filtering, sorting, and data mapping** capabilities.
- Implemented **Progressive Web Application (PWA)** features to enhance performance and user experience.
- Streamlined audio/video file transfers to AWS S3, **preventing web wait timeout issues**.
- Migrated** legacy monolithic applications to microservices architecture hosted in AWS EC2 using AWS API Gateway.
- Integrated **OAuth 2.0 authorization** to secure API endpoints and enhance user authentication process.

Tech Stack: Laravel, ASP.Net Core, React.js, jQuery, Entity Framework, AWS, Docker, Puppet, Nginx

Senior Software Developer | *Proshore* Jan 2021 – June 2022

Software Developer | *Proshore* Mar 2019 – Jan 2021

- Led development of B2B web applications in **cross-functional Agile team of 7-10 members**.
- Achieved **400% performance boost** with targeted **database optimization** and codebase enhancement.
- Optimized large data transformation and integration in Python with factory pattern and **multi-threading** capabilities.
- Optimized SQL queries**, re-indexed tables, implemented client-server database model and horizontal scaling with **database partitioning** and sharding to reduce server load and improve performance.
- Reduced load times by 30% with **lazy-loading and write-through caching strategy** on Redis ElastiCache.
- Automated e-commerce **data synchronization** across 4 vendor systems for improved data accuracy and efficiency.
- Engineered **CI/CD pipeline** reducing deployment time by 70% with improved release and rollback process.

Tech Stack: Node.js, TypeScript, Python, C#, ASP.Net Core, JavaScript, Express.js, Flask, ASP.Net Core, MySQL, AWS, Docker, Kubernetes, Redis, jQuery, Vue.js, Knex.js, Mollie API, Filezilla, FTP, SMTP, cXML, Jira, Postman, Selenium

Associate Software Engineer | *GrowByData Services, Nepal* Aug 2018 – Feb 2019

- Implemented **search engine optimization (SEO)** strategies resulting in increased visibility of e-commerce stores.
- Incorporated search engine spidering, crawling, indexing, and set up campaigns for **A/B testing**.

Tech Stack: JavaScript, PHP, jQuery, Shopify, Magento, WordPress, ABTasty, Screaming Frog, Google Analytics

EDUCATION

Master of Science in Computer Science | *Miami University, Oxford, OH* | CGPA: 4.00 Aug 2022 – Aug 2024
Publication: A case study on Blockchain-based Anonymous Reviewer Incentive Token [IEEE SMC 2024]
Bachelor of Engineering in Computer Engineering | *Kathmandu University* | CGPA: 3.63 Aug 2014 – Oct 2018

PROJECTS

Review Rewards | *Solidity, TypeScript, React.js, OracleDB, Python, Hardhat, OpenGSN, Foundry* Jan 2023 – July 2024

- Developed and deployed secure, privacy-centric web3 peer review platform on Ethereum. Implemented Soul Bound Tokens (non-transferable NFTs) for perpetual recognition and ERC-20 tokens for incentives.

SuggestAbook | *Python, Flask, React.js, MongoDB, Numpy, Pandas* May 2020 – Apr 2023

- Literary journal platform to keep a record of books read, memorable quotes, and relatable character experiences. Implemented personalized book recommendations using k-Nearest Neighbor algorithm on user-item sparse matrix.

LEADERSHIP AND CERTIFICATIONS

ICAgile Certified Professional Apr 2024
Women Leaders in Technology (WLIT) | *Fellowship Technical Trainer* Aug 2018 – June 2022

- Conducted web development sessions in Node.js, JavaScript, and jQuery at WLIT bootcamp 2020/21 and 2021/22.
- Mentored undergraduate female students, guiding them toward their professional aspirations and nurturing leadership abilities.

REFERENCES

References available upon request.